MILITARY SPECIFICATION

PLOTTING BOARD, FLASH RANGING, FIRE CONTROL: M18

1. SCOPE

1.1 This specification covers Flash Ranging Plotting Board M18 which is used to plot the location of enemy batteries (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F8567955	Plotting Board, Flash Ranging, Fire Control: M18

FSC 1220

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PACKAGING DATA SHEET

11728160 Packaging of Plotting Board, Flash Ranging, Fire Control: M18 W/Equipment.

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The plotting board shall be manufactured in accordance with Drawing F8567955 and the respective drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 Storage temperatures.- The plotting boards shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of between 60° and 90°F. after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F.

3.4.2 Operating temperatures.- The plotting boards shall meet the applicable requirements of 3.8 while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.5 Plotting scale flatness.- With the plotting board arranged for two (2) man operation, the plotting scale of the drafting machine shall lie flat on any part of the disk from the "7" graduation to the "15" graduation within .015 inch when the "11" graduation is at the center of the board. The ends of the scale shall lie flat within 0.025 inch.

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3.6 Parallelism.- When alined with the fiducial edge of the plotting scale, the Grid lines of the plotting disk shall be parallel to the fiducial edge of the scale within 0.015 inch throughout their entire length with the azimuth scale and vernier reading 0 ± 0.1 mil and 1600 ± 0.1 mil.

3.7 Accuracy.- Lines drawn forming a centrally located square with sides measuring 15,000 ± 15 meters shall have both diagonals 21,213 ± 21 meters.

3.8 Operability.

3.8.1 Torque.- The torque required to rotate the slow motion knob shall not exceed 9 pound inches at a standard ambient temperature of between 60° and 90°F. and 12 pound inches at the extreme operating temperatures specified in 3.4.2.

3.8.2 Plotting board table.- With the brake lever pressed and held, it shall be possible to rotate the plotting board table through its entire travel without binding. With the brake lever released, the braking mechanism shall hold the table in a clamped position but allow fine adjustment with the slow motion knob.

3.8.3 Braking force.- With the brake lever in its normal position (release) the table shall not move with respect to the vernier scale when a tangential force of 10 pounds is applied at the outer edge of the clamping surface.

3.8.4 Drafting machine.- With the plotting board arranged in both the two man and one man operation mode, there shall be no binding in any part of the drafting machine when the plotting scale of the machine is moved back and forth over the entire plotting disk.

3.8.5 Plotting disk surface.- The plotting disk shall have a surface that will allow clear pencil markings and clean erasures. The surface shall have no defects such as pits, digs, scratches, and stains which may be mistaken for pencil marks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 boards plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table I herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after

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first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I

(Conducted at a temperature between 60° to 90°F)

CLASSIFICATION	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Plotting scale flatness	3.5	4.6.3
102. Parallelism	3.6	4.6.4
103. Accuracy	3.7	4.6.5
104. Torque	3.8.1	4.6.6
105. Plotting board table	3.8.2	Tactile
106. Braking force	3.8.3	4.6.7
107. Drafting machine	3.8.4	Tactile
108. Plotting disk surface	3.8.5	Visual
MINOR: NONE		

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One plotting board shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Materials	3.1	Drawings and Specifications
Fabrication	3.2	Visual and Drawings
General specifications	3.3	Visual and MIL-F-13926

4.4.2 Temperature tests.- Three plotting boards shall be selected at random from each 50 produced, or from each month's production, whichever occurs first, to determine compliance with the storage and operating temperatures as specified in 3.4.1, 3.4.2, 4.6.1 and 4.6.2 respectively.

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by

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the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The storage temperature test for the plotting boards at minus 80 degrees and plus 160 degrees Fahrenheit shall be conducted in conformance with 3.4.1 and the "Test Facilities" of MIL-F-13926. Upon return to standard ambient temperature the plotting boards shall be subjected to a visual and tactile examination and shall show no evidence of physical failure such as cracks, splits, or warpage. The plotting boards shall then be subjected to all the tests specified herein.

4.6.2 Operating temperature.- The temperature cycling for the operability tests of the plotting boards at the specified temperature extremes shall be conducted in accordance with 3.4.2 and the "Test Facilities" of MIL-F-13926. When the plotting boards are thermally stabilized at minus 40 degrees and plus 150 degrees Fahrenheit, respectively, they shall be subjected to the operability tests specified in 4.6.6.

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4.6.3 Plotting scale flatness.- This test shall be performed by placing the "11" graduation of the plotting scale at the center of the board, and measuring any variation in flatness at the "7" and "15" graduations of the scale with standard thickness gages. Any error in flatness shall not exceed the requirements of 3.5.

4.6.4 Parallelism.- Parallelism of the grid lines in relation to the fiducial edge of the plotting scale shall be tested on the two grid lines which intersect through the center of the plotting disc. (Rotate the table 1600 mils to obtain alinement of the perpendicular grid line with the plotting scale.) Use a pocket optical comparator to check errors in parallelism. Errors shall not exceed the requirements of 3.6.

4.6.5 Accuracy.- The plotting board shall be arranged for two man operation with the plotting scale and grid lines parallel and a zero setting on the azimuth scale. With this setting the following procedure shall be utilized to determine compliance with 3.7:

a. Using the plotting scale, measure off and establish a point 7500 meters north of the approximate center of the disk.

b. From the point established above, measure off and establish a point 15000 meters south.

c. Rotate the plotting disk to 1600 mils and repeat the procedures in "a" and "b" on the new north-south line.

d. With the table set in this position, draw lines through the points established on the east-west line.

e. Rotate the board to zero and draw lines through the points established on the east-west line so as to form a square.

f. Rotate the board to a position where the plotting scale bisects the corners of the square.

g. Both diagonals, as measured on the scale, shall be in compliance with the accuracy specified in 3.7.

NOTE: Use a 2H or softer pencil for drawing or marking on the plotting disk surface. Erase pencil markings with a soft rubber eraser void of abrasives or grit.

4.6.6 Torque.- This test shall be performed using a commercial type inch-pound torque wrench with adapter to accommodate the slow motion knob. Readings with the torque wrench shall be taken in both directions while in motion. The torque required to rotate the slow motion knob at standard ambient temperature and at the extreme temperatures of 3.4.2 shall not exceed the values as specified in 3.8.1.

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4.6.7 Braking force.- A standard spring pull-type scale, or equivalent, and an adapter secured to the clamping surface of the table to permit engagement of the scale shall be utilized to obtain a direct pull of 10 pounds in a direction tangential to the table periphery. This test shall be performed with the brake lever released and in its normal position. Upon application of the 10 pound force there shall be no movement of the table to insure compliance with 3.8.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11728160. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Flash Ranging Plotting Board, M18 is used to determine the location of hostile batteries by plotting the line of azimuth of the flash or smoke from the weapons as reported by two or more observation posts. Plotting Board M18 may also be used for sound ranging through installation of a supplementary scale and index tabs.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable Packaging Data Sheet (see 5.1).
(e) Provision for first article testing.

Custodian:

Army - MU

Preparing activity:

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Project No. 1220-A228

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